Zeroth-Order Optimization

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Contents

Chapter 1

Accelerated Zeroth-Order Mirror Descent via Linear Mixing

Assumption 1.1. We assume that the smoothness constant of f is $L \geq 1$. Otherwise, we can replace it with 1. Also assume that $0 < \delta < 1/e$.

We use the linear mixing framework to design an accelerated zero-order gradient descent algorithm. The idea is to query gradient descent y_t and mirror descent z_t at the same point x_t, and then linearly mix the answers $x_{t+1} = \lambda z_t + (1-\lambda) y_t$. The algorithm can be described as

$$\begin{aligned}
x_{t+1} &= \lambda y_t + (1-\lambda) z_t \\
y_{t+1} &= x_{t+1} - \eta \tilde{\nabla} f(x_{t+1}) \\
z_{t+1} &= \mathrm{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1}))
\end{aligned}$$

The first thing to do is to deal with quadratic error.

Lemma 1.2 (Quadratic Error)**.**

$$\left\| \tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^{r} p_i p_i^\top \nabla f(x) \right\| \leq L\sigma \frac{1}{r} \sum_{i=1}^{r} \|p_i\|^3$$

Proof. We have

$$\begin{aligned}
\frac{f(x+\sigma p) - f(x-\sigma p)}{2\sigma} p - pp^\top \nabla f(x) &= \frac{1}{2\sigma} \left[\int_{-1}^{1} \frac{\mathrm{d}}{\mathrm{d}t} f(x + t\sigma p) - \sigma pp^\top \nabla f(x) dt \right] \\
&= \frac{1}{2\sigma} \left[\int_{-1}^{1} \langle \nabla f(x + t\sigma p) - \nabla f(x), \sigma p \rangle p dt \right]
\end{aligned}$$

Now, by L-smoothness,

$$\left| \int_{-1}^{1} \langle \nabla f(x + \sigma t p) - \nabla f(x), \sigma p \rangle p dt \right| \leq \int_{-1}^{1} L\sigma^2 |t| \, \|p\|^3 \, dt$$

Therefore,

$$\left|\frac{f(x+\sigma p)-f(x-\sigma p)}{2\sigma}p - pp^\top \nabla f(x)\right| \leq L\sigma \left\|p\right\|^3$$

Then by triangle inequality,

$$\left\|\tilde{\nabla} f(x) - \frac{1}{r}\sum_{i=1}^{r} p_i p_i^\top \nabla f(x)\right\| \leq L\sigma \frac{1}{r}\sum_{i=1}^{r} \left\|p_i\right\|^3.$$

$\square$

Unfortunately, this bound is necessary. Ideally we would like to just say that $f(x+\sigma p)-f(x-\sigma p)/2\sigma p = pp^\top \nabla g(x)$ for some function $g(x)$ related to $f(x)$. Possibly a smoothed variant f_σ. But this cannot possible hold for every p unless f is quadratic in a neighborhood of x. To see this, we can use MVT to expand the LHS as $\langle \nabla f(x+\theta\sigma p), p\rangle p$. By taking $p = e_i$, we see that $e_i^\top \nabla g(x) = e_i^\top \nabla f(x+\theta_i \sigma e_i)$. Sending $\sigma \to 0$, $\nabla f(x+\theta_i\sigma e_i) \to \nabla f(x)$ under L-smoothness assumption, meaning that $\nabla g(x) = \nabla f(x)$. This would say that f is quadratic, which is not true in general.

1.1 Gradient Progress

Recall the following lemma from before.

Lemma 1.3. *Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d, with*

$$A \coloneqq \left\{dL_\delta^2 \leq X \leq 3d\log(C/\delta)\right\}$$

we have

$$\mathbb{E}[(X-d); A] = 0, \quad \mathbb{P}(A) \geq 1-\delta$$

for some absolute constant C.

These lemmas tell us how much progress the gradient descent step is making.

Lemma 1.4 (Vershynin (2018) Corollary 7.3.2)**.** *Let $P \in \mathbb{R}^{d\times r}$ be a matrix with i.i.d. Gaussian entries. Define $\left\|A\right\|_{\mathrm{op}} = \sup_{u\in\mathbb{S}^{r-1}} \left\|Au\right\|_2$. Then*

$$\mathbb{P}\Big(\left\|P\right\|_{\mathrm{op}} \geq \sqrt{d}+\sqrt{r}+s\Big) \leq 2e^{-cs^2}$$

Proposition 1.5. *Define $\chi = \log(CrT/\delta)$ and let $P_t = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r}\end{bmatrix}$. Define the bounded operator norm event*

$$\mathfrak{O}_t = \left\{\frac{1}{\sqrt{r}}\left\|P\right\|_{\mathrm{op}} \leq C\frac{d}{r}\sqrt{\log(cTr/\delta)}\right\}$$

Then on $\mathfrak{O}_t$, for every $0 \le t \le T-1$,

$$\left\| \frac{1}{r} \sum_{i=1}^{r} p_{t,i} p_{t,i}^\top \nabla f(x_t) \right\|^2 \le \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^{r} |p_{t,i}^\top \nabla f(x_t)|^2$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{O}_t \right) \ge 1 - \delta$$

Proof. Write $P = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r} \end{bmatrix} \in \mathbb{R}^{d \times r}$. By choosing $s = \sqrt{1/c \log(2/\delta)}$ above, we know that

$$\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \le 1 + \sqrt{\frac{d}{r}} + \frac{1}{c\sqrt{r}} \sqrt{\log(c/\delta)} \le C \sqrt{\frac{d}{r}} \sqrt{\log(c/\delta)}$$

with probability at least $1-\delta$. As $PP^\top = \sum_{i=1}^r p_{t,i} p_{t,i}^\top$, we see that

$$\left\| \frac{1}{r} PP^\top \nabla f(x_t) \right\|^2 \le \left(\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \right)^2 \frac{1}{r} \left\| P^\top \nabla f(x_t) \right\|^2 \le \frac{Cd}{r} \log(c/\delta) \frac{1}{r} \sum_{i=1}^{r} |p_{t,i}^\top \nabla f(x_t)|^2$$

We then rescale δ to δ/Tr to complete the proof. □

Lemma 1.6. *Define the big progress event*

$$\mathfrak{P}_t = \left\{ \frac{1}{r} \sum_{i=1}^{r} |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \ge \frac{1}{2} \|\nabla f(x_t)\|^2 \right\}$$

As long as $r \ge 16\chi$, we have

$$\mathbb{P}\Big(\bigcap \mathfrak{P}_t \Big) \ge 1 - \delta.$$

Proof. As $\langle p_{t,i}, \nabla f(x_t) \rangle \overset{i.i.d.}{\sim} \mathcal{N}(0, \|\nabla f(x_t)\|^2)$, by the Laurent-Massart inequality (Laurent and Massart, 2000, Lemma 1, Equation (4.4)),

$$\mathbb{P}\Big(\chi_r^2 \le \frac{r}{2} \Big) \le \exp\Big(-\frac{r}{16} \Big)$$

Therefore as long as $r \ge 16\chi \ge 16 \log(T/\delta)$,

$$\frac{1}{r} \sum_{i=1}^{r} |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \ge \frac{1}{2} \|\nabla f(x_t)\|^2$$

for all $0 \le t \le T-1$. □

Discussion. It is unfortunate to necessitate a lower bound on r which will eventually grow with ε. The way around this is usually to instead look at the gradient progress over a time interval I of length $\asymp \log(T/\delta)$, in which case we could apply the same bound as above. However, this would require the gradients to be of roughly the same size, otherwise one gradient could dominate and we would only really have r effective samples even over the whole interval (compared to $r\log(T/\delta)$).

When proving gradient descent converges to a first-order stationary point (in the sequel), this is possible, and alleviates this requirement on r. However, in the mixing case, since the z_t point depends only on z_{t-1} and not on y_{t-1}, the mirrored sequence could drift extremely far from the gradient sequence. In this case, $x_{t+1} - x_t = (1-\lambda)(y_t - x_t) + \lambda z_t - \lambda x_t$ could potentially grow unbounded. For example, consider the one-variable function $f(x) = Gx + \frac{L}{2}x^2$. Assume that $x_t = 0$, we know that $\nabla f(x_t) = G$. Then $y_t = -\eta G$. $z_t = z_{t-1} - \alpha G$. Then

$$|x_{t+1} - x_t| = |x_{t+1}| = |(1-\lambda) - \eta G + \lambda(z_{t-1} - \alpha G)|$$

Sending $|z_{t-1}| \to \infty$, this gap can become unbounded. The α dictating z_{t-1}'s progression grows with ε, so this is (a priori) not an unrealistic scenario.

Lemma 1.7. *Define the bounded-perturbation event*

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^{r} \left\{ \sqrt{dL_{\delta/rT}} \le \|p_{t+1,i}\| \le \sqrt{3d\chi} \right\}$$

Then on $\mathfrak{B}_{t+1}$, as long as $\eta \le r/(12Ld\chi)$, we have

$$f(y_{t+1}) \le f(x_{t+1}) - \frac{\eta}{8}\|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2\chi^3 d^3$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{B}_t\right) \ge 1 - \delta$$

Proof. By L-smoothness,

$$\begin{aligned} f(y_{t+1}) &\le f(x_{t+1}) + \langle \nabla f(x_{t+1}), y_{t+1} - x_{t+1}\rangle + \frac{L}{2}\|y_{t+1} - x_{t+1}\|^2 \\ &= f(x_{t+1}) - \eta\langle \nabla f(x_{t+1}), \tilde{\nabla} f(x_{t+1})\rangle + \frac{L\eta^2}{2}\left\|\tilde{\nabla} f(x_{t+1})\right\|^2 \end{aligned}$$

Writing $\tilde{\nabla} f(x_{t+1}) = \frac{1}{r}\sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) + R_{t+1}$, we get

$$\begin{aligned} f(y_{t+1}) \le f(x_{t+1}) &- \frac{\eta}{r}\sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 - \eta\langle \nabla f(x_{t+1}), R_{t+1}\rangle \\ &+ L\eta^2 \left\| \frac{1}{r}\sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1})\right\|^2 + L\eta^2\|R_{t+1}\|^2 \end{aligned}$$

Now, $2\langle \eta \nabla f(x_{t+1}), R_{t+1}\rangle \leq \eta^2 \left\|\nabla f(x_{t+1})\right\|^2 + R_{t+1}^2$ by Cauchy-Schwarz and Young's inequality.

Define the event where p_i are bounded:

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^{r} \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

We know that $\mathbb{P}(\mathfrak{B}_{t+1}) \geq 1 - \delta/rT$. Thus,

$$\begin{aligned} f(y_{t+1}) \leq f(x_{t+1}) &- \frac{\eta}{r}\sum_{i=1}^{r} |p_i^\top \nabla f(x_{t+1})|^2 + \eta^2\gamma \left\|\nabla f(x_{t+1})\right\|^2 + (L\eta^2 + \gamma)\left\|R_{t+1}\right\|^2 \\ &+ \frac{CLd\eta^2 \log(Tr/\delta)}{r} \cdot \frac{1}{r}\sum_{i=1}^{r} |p_i^\top \nabla f(x_{t+1})|^2 \end{aligned}$$

From quadratic error lemma we know that $|R_{t+1}| \leq 3L\sigma \log^{3/2}(1/\delta) d^{3/2}$. Now by choosing $\eta \leq r/(4CLd\chi)$, we know that $-\eta + CLd\chi\eta^2/r \leq -3/4\eta$. As $L\eta^2 < 1$ as $L \geq 1$, possibly changing the absolute constant C', we get,

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{3\eta}{4}\frac{1}{r}\sum_{i=1}^{r} |p_i^\top \nabla f(x_{t+1})|^2 + \eta^2 \left\|\nabla f(x_{t+1})\right\|^2 + C'L^2\sigma^2\chi^3 d^3$$

As long as $\eta \leq 1/8$, $-3\eta/8 + \eta^2 \leq -\eta/4$, so using the previous lemma, we ensure

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{\eta}{8}\left\|\nabla f(x_{t+1})\right\|^2 + C'L^2\sigma^2\chi^3 d^3$$

□

1.2 Mirror Descent against Progress

We can copy the mirror descent lemma from last time:

Lemma 1.8. *If $z_{t+1} = \mathrm{Mirr}_{z_t}(\alpha\tilde{\nabla} f(x_{t+1}))$, then*

$$\forall u \in \mathbb{R}^d \quad \alpha\langle \tilde{\nabla} f(x_{t+1}), z_t - u\rangle \leq \frac{\alpha^2}{2}\left\|\tilde{\nabla} f(x_{t+1})\right\|^2 + V_{z_t}(u) - V_{z_{t+1}}(u).$$

The standard coupling lemma comes from using $\left\|\nabla f(x_{t+1})\right\|^2 \leq 2L(f(x_{t+1}) - f(y_{t+1}))$. However this doesn't work here, as we are using an estimator. But we can fix that using the following lemma.

Lemma 1.9. *Define the event*

$$\mathfrak{G}_{t+1} = \left\{ \left\| \frac{1}{r}\sum_{i=1}^{r} p_{t+1,i}p_{t+1,i}^\top \nabla f(x_{t+1}) \right\|^2 \leq \frac{Cd}{r}\chi \left\|\nabla f(x_{t+1})\right\|^2 \right\}$$

On $\mathfrak{B}_{t+1} \cap \mathfrak{G}_{t+1} \cap \mathfrak{O}_{t+1} \cap \mathfrak{P}_{t+1}$, we have for all slack $\gamma \geq 2$,

$$\frac{Cd\gamma\chi}{r} \|\nabla f(x_{t+1})\|^2 + \left\|\tilde{\nabla} f(x_{t+1})\right\|^2 \leq \frac{16Cd\gamma\chi}{r\eta} \left(f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2\chi^3 d^3\right)$$

Furthermore,

$$\mathbb{P}\Big(\bigcap \mathfrak{G}_t\Big) \geq 1 - \delta.$$

Proof. Define the event

$$\mathfrak{G}_{t+1} = \left\{ \left\| \frac{1}{r} \sum_{i=1}^{r} p_{t+1,i} p_{t+1,i}^\top \nabla f(x_{t+1}) \right\|^2 \leq \frac{Cd}{r} \chi \|\nabla f(x_{t+1})\|^2 \right\}$$

Then on $\mathfrak{G}_{t+1}$,

$$\begin{aligned} \left\|\tilde{\nabla} f(x_{t+1})\right\|^2 &\leq 2 \left\| \frac{1}{r} \sum_{i=1}^{r} p_i p_i^\top \nabla f(x_{t+1}) \right\|^2 + 2\|R_{t+1}\|^2 \\ &\leq 2\frac{Cd}{r}\chi \cdot \|\nabla f(x_{t+1})\|^2 + 2\|R_{t+1}\|^2 \\ &\leq \frac{2Cd}{r}\chi \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3 \end{aligned}$$

We will need some slack of the gradient norm to tune later. For $\gamma \geq 2$, as $\gamma - \gamma/2 \geq 1$, this is at most

$$\frac{2Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3 - \frac{Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2$$

From before we know that

$$\begin{aligned} &\frac{\eta}{8} \|\nabla f(x_{t+1})\|^2 \\ &\leq f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2 \log^3(rT/\delta) d^3 \end{aligned}$$

Therefore an upper bound on $\left\|\tilde{\nabla} f(x_{t+1})\right\|^2$ is

$$\begin{aligned} &\frac{16Cd\gamma\chi}{r\eta} \left[f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2 \log^3(rT/\delta) d^3\right] \\ &+ C'L^2\sigma^2\chi^3 d^3 - \frac{Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2 \end{aligned}$$

As $\eta < 1/8$ we can absorb the last term into the first by changing the constant C'. Finally, after an application of the triangle inequality and Lemma **??**, we have $\mathbb{P}(\mathfrak{G}_{t+1}) \geq 1 - \delta/T$, which gives the statement of the lemma. □

1.3 Mirror Descent and Linear Mixing

This lemma is the same as before.

Lemma 1.10. *When λ is chosen so $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$, the "regret" is bounded by*

$$\alpha\langle\nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle\nabla f(x_{t+1}), z_t - u\rangle \le \frac{8Cd\gamma\chi\alpha^2}{\eta}(f(y_t) - f(x_{t+1}))$$

Proof. We have by convexity and the definition of $x_{t+1} = \lambda z_t + (1-\lambda)y_t$,

$$\begin{aligned}
&\alpha\langle\nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle\nabla f(x_{t+1}), z_t - u\rangle \\
&= \alpha\langle\nabla f(x_{t+1}), x_{t+1} - z_t\rangle = \frac{(1-\lambda)\alpha}{\lambda}\langle\nabla f(x_{t+1}), y_t - x_{t+1}\rangle \\
&\le \frac{(1-\lambda)\alpha}{\lambda}(f(y_t) - f(x_{t+1}))
\end{aligned}$$

Now let $\lambda \in (0,1)$ satisfy $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$ and we are done. □

This lemma reveals the martingale term we need to bound.

Lemma 1.11.

$$\begin{aligned}
\alpha\langle\nabla f(x_{t+1}), z_t - u\rangle \le \alpha\langle\tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \alpha\sum_{i=1}^{r}\langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle \\
+ C'T\alpha^2\left\|R_{t+1}\right\|^2 + \frac{1}{4T}V_{z_t}(u).
\end{aligned}$$

Proof.

$$\langle\nabla f(x_{t+1}), z_t - u\rangle = \langle\tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \sum_{i=1}^{r}\langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle + \langle R_{t+1}, z_t - u\rangle$$

Now, by Young's inequality, $\alpha\langle R_{t+1}, z_t - u\rangle \le C'\alpha^2 T\left\|R_{t+1}\right\|^2 + \frac{1}{8T}\left\|z_t - u\right\|^2$, by strong convexity of Bregman divergence, $V_{z_t}(u) \ge \frac{1}{2}\left\|z_t - u\right\|^2$, so we arrive at the statement of the lemma. □

1.4 Dealing with the Martingale

The main idea from here is that, the martingale term

$$\sum_{t=0}^{T-1}\sum_{i=1}^{r}\langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle$$

can be bounded by the "Energy" $V_{z_t}(x^*)$, which itself can be bounded by the sum of the gradient norms $\sum\left\|\nabla f(x_t)\right\|^2$ and $\Theta \ge V_{z_0}(x^*)$.

This highlights a very general principle. To deal with martingale error terms with high probability, try bounding them by some other part of the equation which should have typical size $\Theta(1)$. In this case, that will be $V_{z_t}(x^*)$ for us.

First, we highlight some intuition. Due to oscillatory cancellation, the typical size of the martingale term is much smaller than any deterministic inequality would suggest. This is because it is a martingale difference sequence.

Definition 1.12 (Martingale Difference Sequence)**.** $\{\xi_t\}_{t\geq 1}$ is a martingale difference sequence wrt filtration $\mathcal{F}_t$ if $\xi_t \in \mathcal{F}_t$ and

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$$

The main intuition is that in a martingale difference sequence, the variance of the full sum is the sum of the variance of each ξ_t, which is vastly superior than including all the cross covariance terms $\mathrm{Cov}(\xi_i, \xi_j)$. This improves our bound substantially.

Lemma 1.13. *Define* $S_T = \sum_{t=1}^T \xi_t$*. Then*

$$\mathbb{E}\left[S_T^2\right] = \sum_{t=1}^T \mathbb{E}\left[\xi_t^2\right]$$

Proof. We have

$$\mathbb{E}\left[M_T^2\right] = \sum \mathbb{E}\left[\xi_t^2\right] + 2\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right]$$

Now,

$$\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right] = \sum_{i<j} \mathbb{E}[\mathbb{E}[\xi_i \xi_j \mid \mathcal{F}_i]] = \sum_{i<j} \mathbb{E}[\xi_i \mathbb{E}[\xi_j \mid \mathcal{F}_i]] = 0.$$

□

Lemma 1.14. *Define*

$$\mathfrak{M}_{t+1} = \left\{ \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^{r} \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \right| \leq \alpha \sqrt{Cr \sum_{t=0}^{\tau-1} \chi \left\| \nabla f(x_t) \right\|^2 V_{z_t}(u)} \right\}$$

Then

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{M}_t \right) \geq 1 - \delta.$$

Proof. From Lemma **??** above, we know that

$$\begin{aligned}&\left|\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\alpha\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle\right|\\&\le C\theta\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\log(CrT/\delta)\,\|\nabla f(x_{t+1})\|^2\,\|z_t-u\|^2+\frac{1}{\theta}\log\left(\frac{CrT}{\delta}\right)\end{aligned}$$

Using $V_{z_t}(u)\ge\frac{1}{2}\|z_t-u\|^2$ and balancing the two terms by carefully choosing θ we arrive at the statement of the lemma. □

1.5 Putting it all Together

First, as in the proof of Mirror Descent, we bound the regret, then $V_{z_t}(x^*)$, then last we will tune α.

Lemma 1.15 (Regret Bound). *On* $\bigcap \mathfrak{B}_t\cap\mathfrak{O}_t\cap\mathfrak{P}_t\cap\mathfrak{G}_t\cap\mathfrak{M}_t$*, we have*

$$\begin{aligned}&\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_t),x_t-x^*\rangle\\&\le\frac{8Cd\gamma\chi}{r\eta}\alpha^2(f(y_0)-f(y_\tau))-\frac{C\alpha^2d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2\\&+\alpha\sqrt{\frac{C}{r}\sum_{t=0}^{\tau-1}\chi\,\|\nabla f(x_{t+1})\|^2\,V_{z_t}(x^*)+\frac{C'(\alpha^2T+\alpha T^2)\gamma L^2\sigma^2\chi^4d^4}{r\eta}}\\&+V_{z_0}(x^*)-V_{z_\tau}(x^*)+\frac{1}{4T}\sum_{t=0}^{\tau-1}V_{z_t}(x^*)\end{aligned}$$

Proof. By Lemma 1.10, we know that

$$\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_t),x_t-x^*\rangle\le\sum_{t=0}^{\tau-1}\frac{8Cd\gamma\chi\alpha^2}{\eta}(f(y_t)-f(x_{t+1}))+\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_{t+1}),z_t-u\rangle$$

By Lemma 1.11, we know that

$$\begin{aligned}&\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_{t+1}),z_t-u\rangle\\&\le\alpha\sum_{t=0}^{\tau-1}\left[\langle\tilde\nabla f(x_{t+1}),z_t-u\rangle+\frac{1}{r}\sum_{i=1}^{r}\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle+2T\,\|R_{t+1}\|^2+\frac{1}{T}V_{z_t}(u)\right]\end{aligned}$$

From above we know that

$$\alpha\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle\le\alpha\sqrt{Cr\sum_{t=0}^{\tau-1}\chi\,\|\nabla f(x_{t+1})\|^2\,V_{z_t}(u)}$$

From Lemmas 1.8 and 1.9 we know that

$$\begin{aligned}
&\sum_{t=0}^{\tau-1} \alpha\langle \tilde{\nabla} f(x_{t+1}), z_t - u\rangle \\
&\leq \alpha^2 \sum_{t=0}^{\tau-1} \frac{8Cd\gamma\chi}{r\eta}\left(f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2\chi^3 d^3\right) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \\
&+ V_{z_0}(x^*) - V_{z_\tau}(x^*)
\end{aligned}$$

Recall that $\|R_{t+1}\|^2 \leq C'L^2\sigma^2\chi^3 d^3$. Putting all of it together, we get:

$$\begin{aligned}
&\sum_{t=0}^{\tau-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\
&\leq \frac{8Cd\gamma\chi}{r\eta}\alpha^2(f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \\
&+ \alpha\sqrt{\frac{C}{r}\sum_{t=0}^{\tau-1}\chi\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'\alpha^2 T\gamma L^2\sigma^2\chi^4 d^4}{r\eta} + C'\alpha T^2 L^2\sigma^2\chi^3 d^3 \\
&+ V_{z_0}(x^*) - V_{z_\tau}(x^*) + \frac{1}{4T}\sum_{t=0}^{\tau-1} V_{z_t}(x^*)
\end{aligned}$$

By using $r \leq d$ and $\eta < 1$ to handle the two quadratic terms we arrive at the statement of the lemma. □

This next lemma is what we meant by bounding the martingale by other already-present $\Theta(1)$ terms.

Lemma 1.16 (Energy Bound)**.** *On $\bigcap \mathfrak{B}_t \cap \mathfrak{O}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t$, upon defining $M = \sup_{0\leq t\leq T-1} V_{z_t}(x^*)$, choosing the slack $\gamma = 8$ and letting $\Theta \geq V_{z_0}(x^*)$ be any upper bound, we have*

$$\frac{M}{2} \leq \frac{256Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \Theta + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta}$$

Proof. By convexity we know that $f^* = f(x^*) \geq f(x) + \langle \nabla f(x), x^* - x\rangle$. Rearranging, we get $\langle \nabla f(x), x^* - x\rangle \geq f(x) - f^* \geq 0$. Using Young's inequality, we know that

$$\alpha\sqrt{\frac{C}{r}\chi\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} \leq \frac{M}{4} + \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2$$

Therefore by the previous lemma, and using $\frac{1}{T}\sum_{t=0}^{\tau-1} V_{z_\tau}(x^*) \leq \frac{1}{4}M$, we know that

$$V_{z_\tau}(x^*) \leq \frac{8C\gamma d\chi}{r\eta}\alpha^2(f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2$$
$$+ V_{z_0}(x^*) + \frac{M}{4} + \frac{C'(\alpha^2 T + \alpha T^2)\gamma L^2\sigma^2\chi^4 d^4}{r\eta}$$
$$+ \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 + \frac{M}{4}$$

By choosing $\gamma = 8$, as $d \geq 1$, we know that

$$-\frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 + \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \leq 0$$

By using $f(y_0) - f(y_\tau) \leq f(y_0) - f^*$ along with $V_{z_0}(x^*) \leq \Theta$, and taking supremum of both sides, we have

$$M \leq \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \Theta + \frac{M}{2} + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta}$$

which completes the proof. □

Theorem 1.17 (Halving Function Decrease)**.** *Assuming the initial error $f(y_0) - f^* \leq \Delta$, by choosing*

$$\eta = \frac{r}{4CLd\chi}, \qquad \alpha = \frac{r}{16Cd\chi}\sqrt{\frac{\Theta}{L\Delta}}$$

In

$$T = \frac{256Cd\chi}{r}\sqrt{\frac{L\Theta}{\Delta}}$$

iterations we have $f(\bar{x}) - f^* \leq \Delta/2$ with probability at least $1 - 5\delta$.

Proof. On $\bigcap \mathfrak{B}_t \cap \mathfrak{O}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t$ all our previous lemmas hold. Each has probability at least $1 - \delta$ so the intersection of 5 has probability at least $1 - 5\delta$.

By Lemma 1.15, we know that

$$\sum_{t=0}^{T-1}\alpha\langle\nabla f(x_t), x_t - x^*\rangle$$
$$\leq \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0) - f(y_T)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{T-1}\|\nabla f(x_{t+1})\|^2$$
$$+ \alpha\sqrt{\frac{C}{r}\sum_{t=0}^{T-1}\chi\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta}$$
$$+ V_{z_0}(x^*) - V_{z_T}(x^*) + \frac{1}{4T}\sum_{t=0}^{T-1}V_{z_t}(x^*)$$

Using Young's inequality in exactly the way as the previous lemma, we can rearrange this to

$$\begin{aligned}&\sum_{t=0}^{T-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\ &\le \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta} + \Theta + M/4\end{aligned}$$

Using the previous lemma, we have

$$\begin{aligned}&\sum_{t=0}^{T-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\ &\le \frac{128Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta} + 2\Theta\end{aligned}$$

Let $\bar{x} = \sum_{t=0}^{T-1} x_t$. By convexity,

$$T(f(\bar{x}) - f^*) \le \sum_{t=0}^{T-1}(f(x_t) - f^*) \le \sum_{t=0}^{T-1}\langle \nabla f(x_t), x_t - x^*\rangle$$

Therefore, assuming the initial objective gap $f(y_0) - f^* \le \Delta$,

$$f(\bar{x}) - f^* \le \frac{128Cd\chi\alpha\Delta}{r\eta T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^2\sigma^2\chi^4 d^4}{r\eta}$$

Now choose the learning rate $\eta = r/(4CLd\chi)$. Then we get

$$f(\bar{x}) - f^* \le \frac{512C^2 d^2 L\chi^2\alpha\Delta}{r^2 T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^3\sigma^2\chi^5 d^5}{r^2}$$

By choosing

$$\alpha = \frac{r}{16Cd\chi}\sqrt{\frac{\Theta}{L\Delta}},$$

and

$$T = \frac{256Cd\chi}{r}\sqrt{\frac{L\Theta}{\Delta}}$$

as $L \ge 1$ we have $T \ge \alpha$, so the above is at most

$$\frac{\Delta}{4} + \sqrt{\frac{L\Theta}{\Delta}}\frac{C'L^3\sigma^2\chi^6 d^6}{r^3}$$

Then as long as

$$\sigma^2 \leq \frac{r^3 \Delta^{3/2}}{4C' L^{7/2} \chi^6 d^6 \sqrt{\Theta}}$$

We have

$$f(\bar{x}) - f^* \leq \frac{\Delta}{4}$$

thereby halving the objective value. This completes the proof. $\square$

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